

1 Polynomial Arithmetic

1.1 Adding and Subtracting Polynomials

Monomial is an expression that could be a numeral, a variable, or the product of numerals and variables within certain restrictions.

1. Must have whole number exponents.
2. Variables do not appear under simplified radical expressions.
3. Denominators do not contain variables.

Examples: -10 , a , $5t^2$, $3/8x^2y^3$

Binomial is an expression that is the sum of two monomials.

Trinomial is an expression that is the sum of three monomials.

Polynomial is an expression that is a monomial or the sum of two or more monomials.

1.2 Multiplying Polynomials

1. $x^a x^b = x^{a+b}$

Multiplying monomials with the same base you keep the base and add the powers.

2. $(x^a)^b = x^{ab}$

Keep the base and multiply the powers when finding powers of a base.

3. $(xy)^a = x^a y^a$

Raise each factor in the product to the power when finding the powers of a product

1.3 Special Products of Binomials

Conjugates are two binomials with the same two terms but opposite signs separating the terms. They are conjugates of each other.

Example: $3x + 2$ & $3x - 2$

Difference of Squares is the *product* of conjugates which produces a special pattern.

Example: $(x + y)(x - y) = x^2 - y^2$

Square Trinomial is the pattern produced by squaring a binomial.

Example: $(x + y)^2 = x^2 + 2xy + y^2$ & $(x - y)^2 = x^2 - 2xy + y^2$

Notice how only the *first* sign is changed when subtracting a squared binomial.

1.4 Dividing Polynomials

$$a^m/a^n = a^{m-n}$$

This works as long as a does not equal zero.

$$a^{-n} = 1/a^n \text{ \& } 1/a^{-n} = a^n$$

This works as long as a does not equal zero.
Also works in reverse.

1.5 synthetic Division

Synthetic Division is a shortcut for polynomial division when the divisor is the form of $x-a$. Only the coefficients are used when dividing.

Example: Divide $(3x^2 + 2x - 11)$ by $(x - 3)$

First step would be multiplying $(x - 3)$ by $3x^2$ giving you $3x^3 - 9x^2$. You subtract this and keep repeating until you are left with a remainder. The remainder is added at the end by being divided by the divisor.